Bilkent University
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Lab Report 07
06 April 2026
EE102 - 01

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Finite State Machine Implementation Using Discrete Logic Gates

Abstract: The objective of this laboratory session was to design and implement a finite state machine (FSM) on a solderless breadboard using discrete 74-series logic ICs. A four-state synchronous Moore FSM was designed with the state sequence 00 → 01 → 11 → 10 → 00, forming a 2-bit Gray code counter. The next-state logic was derived as D1 = Q0 and D0 = NOT(Q1), requiring only a single 74HC74 dual D flip-flop and one gate of a 74HC04 hex inverter. A function generator provided the clock source, and two LEDs displayed the current state in real time. The circuit was verified on the breadboard by observing the correct four-state cycling sequence on the LED outputs.

1. Introduction

Finite state machines are fundamental building blocks in digital systems. They govern sequential behavior in applications from traffic controllers to communication protocols. While modern implementations typically reside within FPGAs or microcontrollers, constructing an FSM from discrete logic gates on a breadboard provides direct insight into the relationship between state transition equations and physical hardware. In this laboratory session, a four-state Moore FSM was designed from first principles: the state transition diagram was drawn, the next-state

equations were derived from the transition table, and the resulting logic was implemented using a 74HC74 dual D flip-flop and a 74HC04 inverter on a small solderless breadboard. A function generator served as the clock source, and two LEDs connected through active-low driver circuits displayed the binary state.

2. Theory

2.1. Moore Finite State Machine

A Moore FSM is a sequential circuit whose outputs depend solely on the present state, not on the current inputs. The machine is characterized by a set of states, a state transition function that maps the current state to the next state on each clock edge, and an output function that maps each state to an output value. In contrast to a Mealy machine, whose outputs can change asynchronously with input changes, a Moore machine's outputs transition only on clock edges, producing glitch-free output signals.

2.2. State Assignment and Transition Table

The FSM in this design cycles through four states in the sequence S0 → S1 → S2 → S3 → S0, with the state assignment S0 = 00, S1 = 01, S2 = 11, S3 = 10. This assignment follows a Gray code pattern in which only one bit changes per transition, minimizing switching activity and reducing the risk of transient glitches on the outputs.

The state transition table is as follows:

Present State (Q1 Q0)	Next State ($Q1^+$ $Q0^+$)
00	01
01	11
11	10
10	00

Since this is an autonomous counter with no external input, the next state depends exclusively on the present state.

2.3. Next-State Equation Derivation

For D flip-flops, the characteristic equation is $Q^+ = D$, meaning the value present at the D input is captured into Q on the next rising clock edge. Therefore, our task reduced to finding Boolean expressions for D1 and D0 as functions of Q1 and Q0 only.

Derivation of D1: Examining the next-state column for $Q1^+$, the values are 0, 1, 1, 0 for present states 00, 01, 11, 10 respectively. Comparing this with the present value of Q0 (0, 1, 1, 0), they match exactly.Therefore:

$$D1 = Q0$$

Derivation of D0: Examining $Q0^+$, the values are 1, 1, 0, 0 for present states 00, 01, 11, 10. Comparing this with the complement of Q1 (1, 1, 0, 0), they match exactly. Therefore:

$$D0 = Q1' = NOT(Q1)$$

These equations can be verified by substitution into each state. For example, starting from state 00: D1 = Q0 = 0 and D0 = NOT(Q1) = NOT(0) = 1, yielding next state 01. From state 01: D1 = 1 and D0 = NOT(0) = 1, yielding 11. From state 11: D1 = 1 and D0 = NOT(1) = 0, yielding 10. From state 10: D1 = 0 and D0 = NOT(1) = 0, yielding 00. All transitions match our intended sequence.

2.4. Output Equations

Since this is a Moore FSM with outputs equal to the state bits, the output equations are simply:

$$Y1 = Q1 \;\; ; \;\; Y0 = Q0$$

The outputs are displayed on LEDs using the active-low LED circuit specified in the laboratory reference materials. In this configuration, the LED is connected from +5V through a 1 kΩ current-limiting resistor to the output pin. The LED illuminates when the driving signal is LOW. Therefore, the LED cathodes are connected to the Q' (complementary) outputs of the flip-flops: when Q = 1, Q' = 0 (LOW), and current flows through the LED, turning it ON.

2.5. D Flip-Flop and Inverter ICs

The 74HC74 is a dual positive-edge-triggered D flip-flop with asynchronous active-low preset and clear inputs. Each flip-flop captures the D input value on the rising edge of the clock. The complementary outputs Q and Q' are both available, which in this design provides the Q0 signal for the D1 = Q0 connection directly, and the Q' outputs for driving the active-low LEDs.
The 74HC04 is a hex inverter containing six independent NOT gates. In this design, a single gate is used to compute NOT(Q1) from the Q1 output of flip-flop B, providing the D0 input to flip-flop A. Although the Q1' output of the 74HC74 could have been used directly to achieve the same logical function without an additional IC, the inverter was included to satisfy the laboratory requirement of using at least one discrete logic gate in the design.

3. Methodology

3.1. Equipment and Components

Hardware: Solderless breadboard (430 tie-points), 5V DC power supply, function generator (square wave clock source), two LEDs, two 1 kΩ resistors. ICs: 1× 74HC74 (dual D flip-flop), 1× 74HC04 (hex inverter).

Wiring: Assorted jumper wires.

3.2. Design Architecture

The system consists of two ICs connected on a single breadboard:

1. *Flip-flop IC (74HC74):* Both flip-flops within the single 74HC74 package are used. Flip-flop A (pins 1–6) stores Q0, and flip-flop B (pins 8–13) stores Q1. Both flip-flops share a common clock signal from the function generator, connected to pin 3 (CLK1) and pin 11 (CLK2). The asynchronous clear inputs (pin 1 and pin 13) are connected together to a manual reset wire. The preset inputs (pin 4 and pin 10) are tied to +5V to disable the preset function. The cross-feedback connections implement the next-state equations: Q0 (pin 5) is connected to D1 (pin 12), realizing D1 = Q0. The output of the inverter is connected to D0 (pin 2), realizing D0 = NOT(Q1).

2. *Inverter IC (74HC04):* Only one gate of the 74HC04 is used. Q1 (pin 9 of the 74HC74) is connected to the inverter input (pin 1 of the 74HC04). The inverter output (pin 2 of the 74HC04) is routed to D0 (pin 2 of the 74HC74). The remaining five gates are unused; their inputs should ideally be tied to ground to prevent floating-input oscillation, though this was not strictly necessary for correct operation at the low clock frequencies used.

3.3. LED Output Circuits

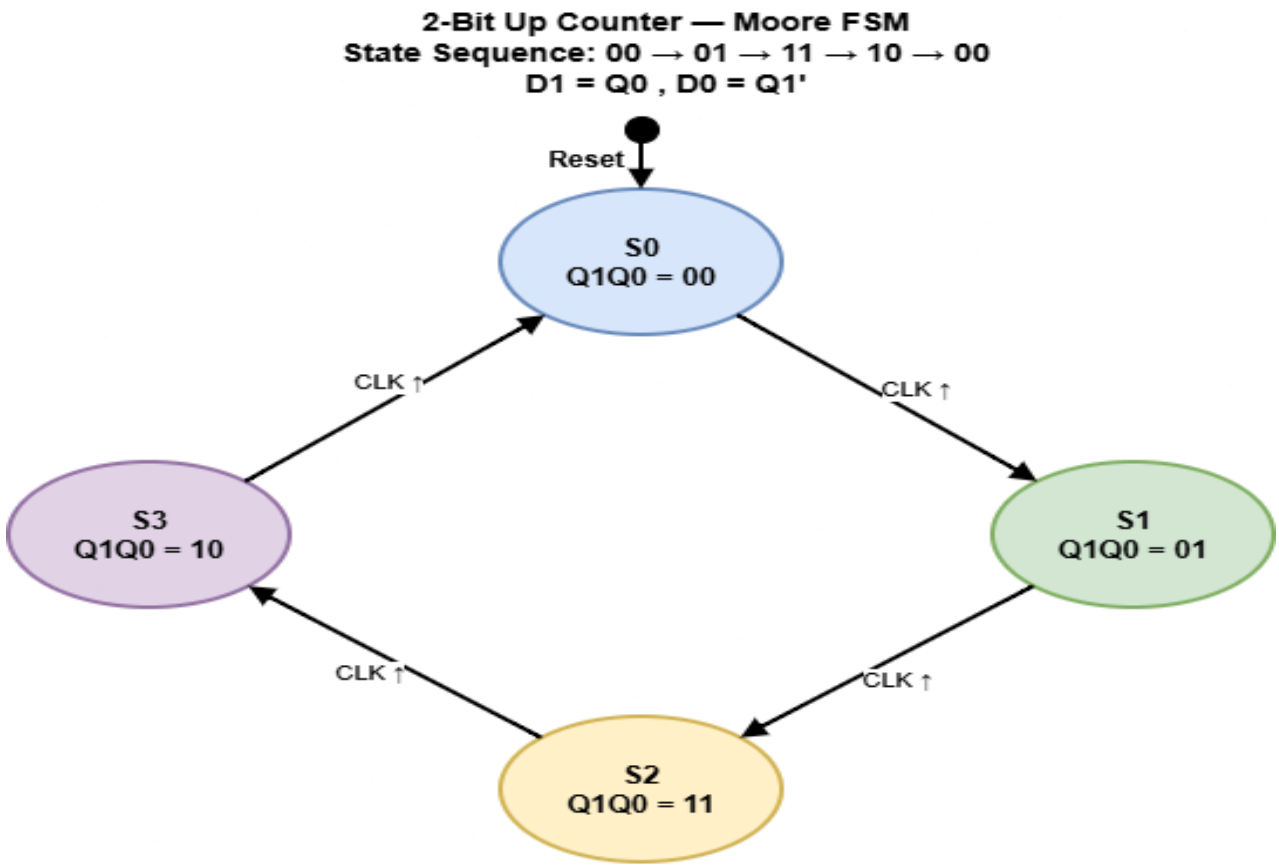
Each LED output follows the active-low circuit specified in the laboratory reference: +5V is connected through a 1 kΩ resistor to the LED anode, and the LED cathode is connected to the Q' output of the corresponding flip-flop. LED0 is connected to Q0' (pin 6) and LED1 is connected to Q1' (pin 8). In this arrangement, an LED turns ON when the corresponding state bit Q is 1 (since Q' = 0, allowing current to flow from +5V through the resistor and LED to the LOW output).

3.4. Clock Source

A function generator was configured to produce a square wave with 5V peak-to-peak amplitude and +2.5V DC offset, resulting in a signal that swings between 0V and 5V. The frequency was initially set to 1 Hz for visual observation of the state transitions on the LEDs. The generator output was connected to both CLK1 (pin 3) and CLK2 (pin 11) of the 74HC74.

3.5. Reset Mechanism

A jumper wire connected to the shared CLR' junction (pins 1 and 13) serves as a manual reset. During normal operation, this wire is held at +5V (clear disabled). To reset the FSM to state S0 (00), the wire is momentarily connected to GND, which asynchronously forces both Q0 and Q1 to 0.



Output Table (Moore — outputs = state bits)

State	Q1	Q0	LED1	LED0
S0	0	0	OFF	OFF
S1	0	1	OFF	ON
S2	1	1	ON	ON
S3	1	0	ON	OFF

Figure 1: State transition diagram showing four states S0(00), S1(01), S2(11), S3(10) with transitions labeled CLK↑ and a reset arrow to S0.

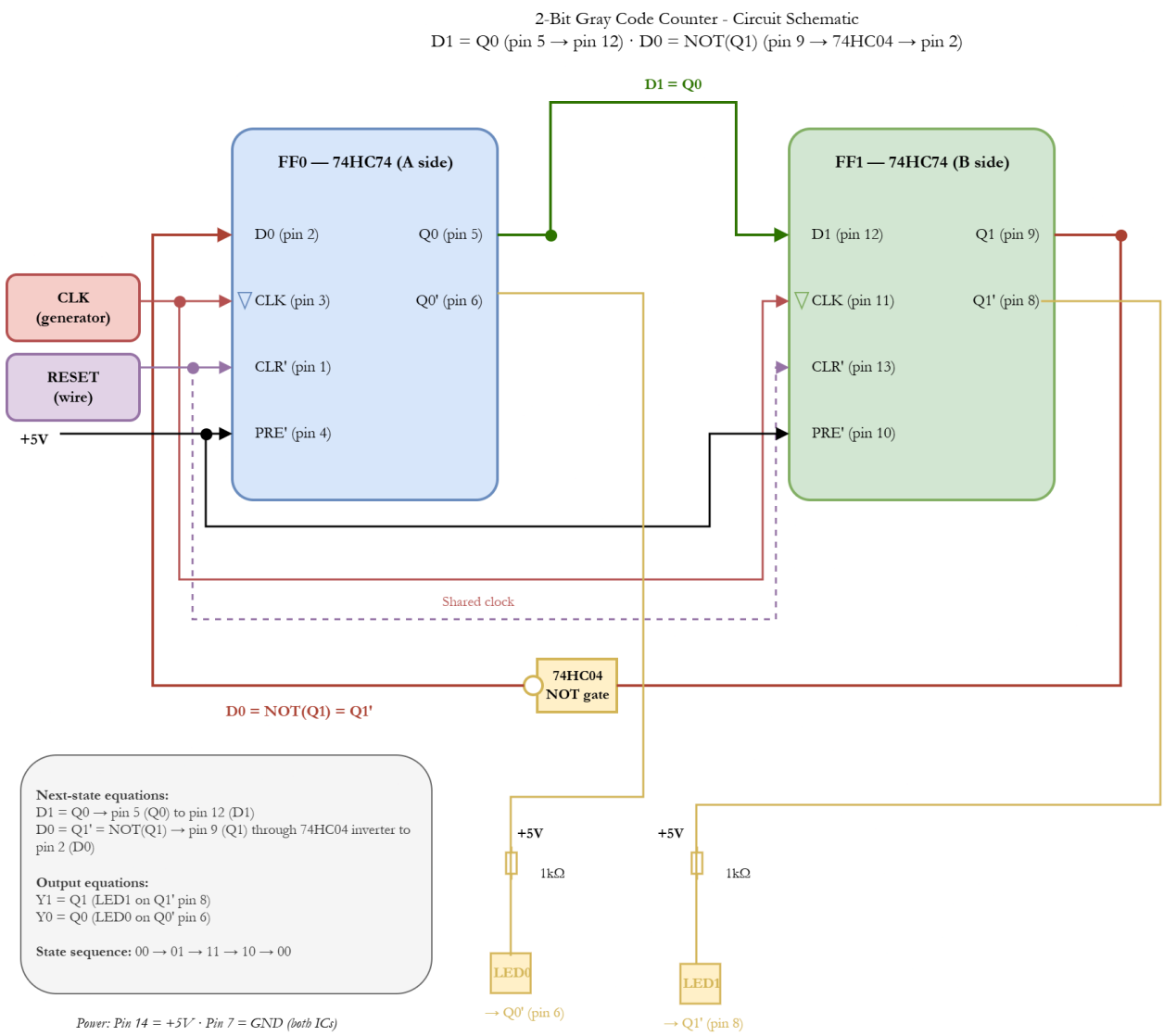


Figure 2: Circuit schematic showing 74HC74, 74HC04, LED circuits, clock input, and reset connections.

4. Results and Analysis

4.1. Functional Verification

After applying power and resetting the FSM, the circuit was verified by observing the LED states at each clock cycle. With the function generator set to 1 Hz, the following sequence was observed repeating continuously:

Clock Cycle	Q1	Q0	LED1	LED0	State
After reset	0	0	OFF	OFF	S0
1st rising edge	0	1	OFF	ON	S1
2nd rising edge	1	1	ON	ON	S2

3rd rising edge	1	0	ON	OFF	S3
4th rising edge	0	0	OFF	OFF	S0

The sequence 00 → 01 → 11 → 10 → 00 repeated indefinitely without error, confirming the correct implementation of our Gray code counter FSM.

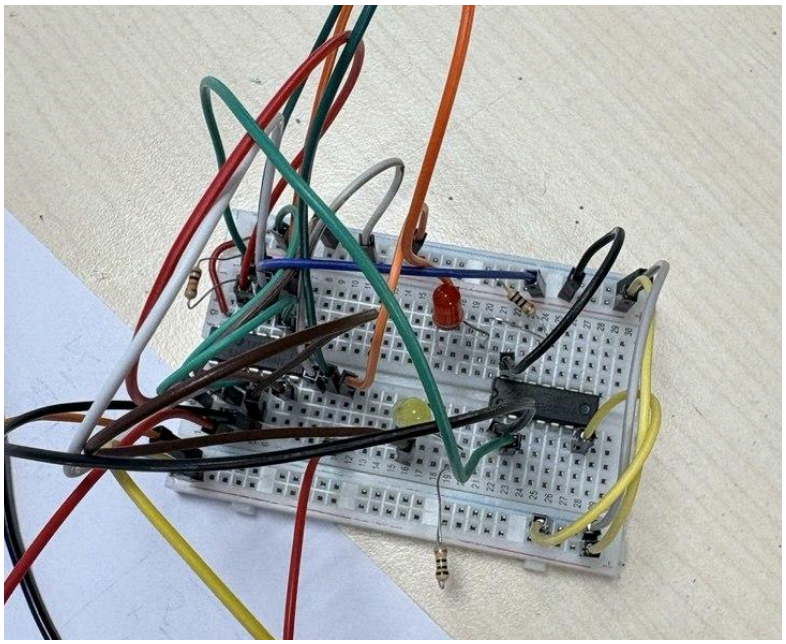


Figure 3: Breadboard photograph showing State S0 (Q1Q0 = 00): both LEDs OFF.

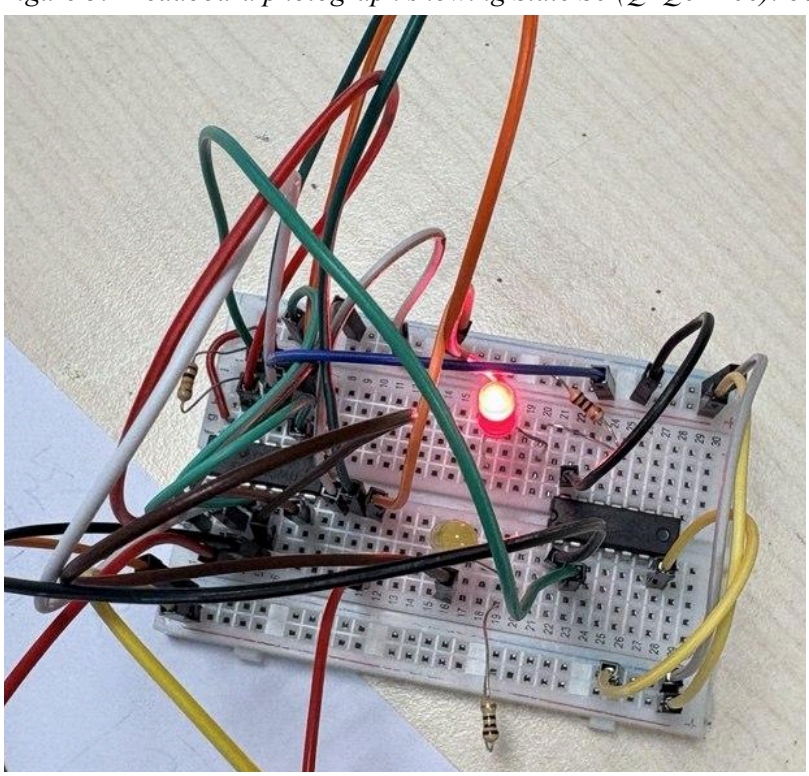


Figure 4: Breadboard photograph showing State S1 (Q1Q0 = 01): LED0 ON, LED1 OFF.

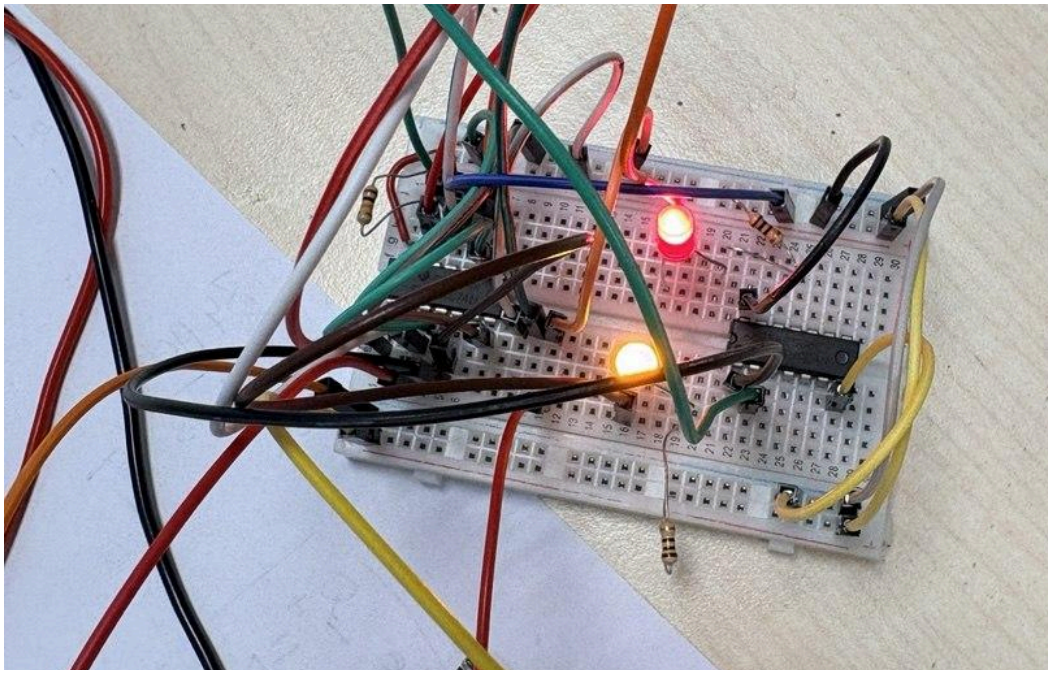


Figure 5: Breadboard photograph showing State S2 (Q1Q0 = 11): both LEDs ON.

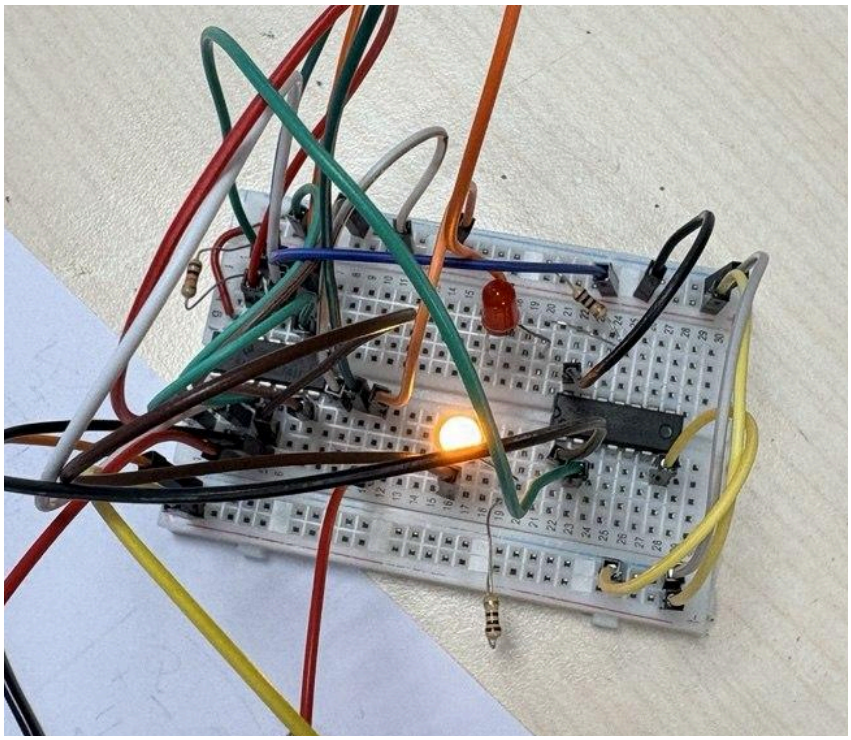


Figure 6: Breadboard photograph showing State S3 (Q1Q0 = 10): LED1 ON, LED0 OFF.

4.2. Reset Verification

The asynchronous reset was tested by momentarily connecting the CLR' wire to GND during various points in the counting sequence. In all cases, both LEDs immediately turned OFF (state 00), and the counter resumed the correct sequence from S0 on the next rising clock edge. This

confirms that the asynchronous clear function of the 74HC74 operates correctly independent of the clock state.

5. Design Considerations

1. *Elimination of Extra Logic Gates:* The chosen state assignment 00 → 01 → 11 → 10 produces next-state equations D1 = Q0 and D0 = Q1' that require minimal combinational logic. The D1 = Q0 connection is a direct wire from one flip-flop output to the other's D input, requiring no gate at all. Only D0 = NOT(Q1) requires a logic gate, and a single inverter from the 74HC04 fulfills this requirement. This minimality arises from the Gray code structure of the state sequence: because only one state variable changes per transition, each next-state equation depends on only one present-state variable. A different state assignment, such as a standard binary count 00 → 01 → 10 → 11, would require an XOR gate for the D1 equation, increasing the IC count.

2. *Synchronous Clocking Architecture:* This design uses a fully synchronous architecture in which both flip-flops receive the same clock signal simultaneously from the function generator. An alternative ripple counter approach, where Q0' drives the clock input of flip-flop B, would eliminate the need for the inverter IC but would introduce clock skew between the two state bits. In the synchronous design, both outputs update at the same instant on the clock edge, ensuring that the state transitions are clean and the LED outputs never display a transient intermediate state.

3. *Active-Low LED Polarity:* The LED circuit topology, in which the LED is driven from the Q' outputs, inverts the visual representation: the LED lights when Q = 1. An initial wiring error in which the LEDs were connected with reversed polarity produced an inverted display pattern. Reversing the LED orientation corrected the issue. This experience highlights the importance of verifying LED polarity during breadboard construction.

4. *CMOS Floating Input Sensitivity:* The 74HC74 uses CMOS technology, which presents very high input impedance. Any input pin left unconnected (floating) can pick up stray electromagnetic noise and oscillate between logic levels. This was observed during initial testing when the clock pin was momentarily disconnected, causing rapid random toggling of the outputs. Ensuring that all active inputs are connected to defined logic levels at all times is essential for reliable CMOS circuit operation.
5. *Justification for the Inverter IC:* The Q1' signal required for D0 is already available at pin 8 of the 74HC74, and a direct wire from pin 8 to pin 2 would have produced the same logical function without any additional IC. The 74HC04 inverter was added explicitly to satisfy the laboratory requirement that at least one discrete logic gate be used in the

implementation. The inverter takes Q1 from pin 9 of the 74HC74, computes its complement, and feeds the result to D0 at pin 2, achieving D0 = NOT(Q1) through an explicit gate rather than through the flip-flop's built-in complementary output.

6. Conclusion

This laboratory session helped me learn the design and physical implementation of a four-state synchronous Moore FSM using discrete ICs. The state sequence 00 → 01 → 11 → 10 → 00 was realized with minimal hardware: a single 74HC74 dual D flip-flop and one gate of a 74HC04 hex inverter. The next-state equations D1 = Q0 and D0 = NOT(Q1) were derived directly from the state transition table and verified both analytically and through hardware testing. The LED outputs confirmed correct state cycling, and the asynchronous reset reliably initialized the machine to state S0. The exercise reinforced the connection between abstract state machine design and physical circuit realization, and highlighted practical considerations including CMOS input sensitivity, LED polarity, and the advantages of synchronous clocking architectures.

Appendix: Pin Connections Summary

Connection	From	To	Function
Power	+5V rail	74HC74 pin 14, 74HC04 pin 14	IC power supply
Ground	GND rail	74HC74 pin 7, 74HC04 pin 7	IC ground
Preset disable	+5V rail	74HC74 pin 4, pin 10	Disable preset (active-low)
Reset	Manual wire	74HC74 pin 1, pin 13	Asynchronous clear (active-low)
Clock	Function generator	74HC74 pin 3, pin 11	Shared clock to both FFs
D1 = Q0	74HC74 pin 5 (Q0)	74HC74 pin 12 (D1)	Next-state feedback
Q1 to inverter	74HC74 pin 9 (Q1)	74HC04 pin 1 (input)	Inverter input
D0 = NOT(Q1)	74HC04 pin 2 (output)	74HC74 pin 2 (D0)	Next-state feedback
LED0	74HC74 pin 6 (Q0')	LED cathode	Active-low LED driver
LED1	74HC74 pin 8 (Q1')	LED cathode	Active-low LED driver